

Deep Learning and Foundation Models for Weather Prediction: A Survey

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Abstract

Physics-based numerical models have been the bedrock of atmospheric sciences for decades, offering robust solutions but often at the cost of significant computational resources. Deep learning (DL) models have emerged as powerful tools in meteorology, capable of analyzing complex weather and climate data by learning intricate dependencies and providing rapid predictions once trained. While these models demonstrate promising performance in weather prediction, often surpassing traditional physics-based methods, they still face critical challenges. This paper presents a comprehensive survey of recent deep learning and foundation models for weather prediction. We propose a taxonomy to classify existing models based on their training paradigms: *deterministic predictive learning*, *probabilistic generative learning*, and *pre-training and fine-tuning*. For each paradigm, we delve into the underlying model architectures, address major challenges, offer key insights, and propose targeted directions for future research. Furthermore, we explore real-world applications of these methods and provide a curated summary of open-source code repositories and widely used datasets, aiming to bridge research advancements with practical implementations while fostering open and trustworthy scientific practices in adopting cutting-edge artificial intelligence for weather prediction. The related sources are available at <https://github.com/JimengShi/DL-Foundation-Models-Weather>.

1 Introduction

Global climate change has increased the frequency of extreme weather events, such as heatwaves, extreme cold spells, intense rainfall, storms, and hurricanes, leading to disasters such as droughts, floods, and air pollution. These changes have profound implications across multiple domains, affecting human health and activities (Flandroy

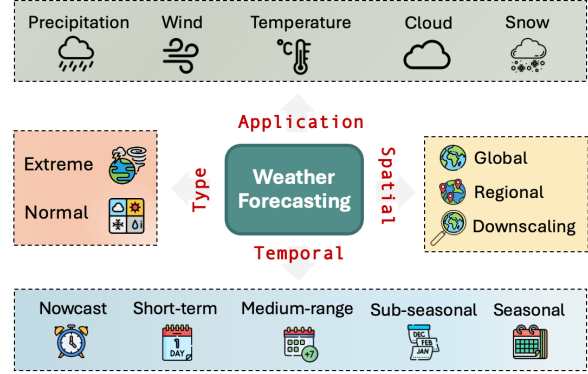


Figure 1: Perspectives of weather forecasting.

et al., 2018), compromising environmental sustainability (Abbass et al., 2022), disrupting economic stability (Carleton and Hsiang, 2016), and altering ecosystem dynamics (Descombes et al., 2020). In this context, developing accurate and timely weather prediction is critical to mitigating these impacts and supporting adaptive strategies.

Physics-based models, including General Circulation Models (GCMs) (Ravindra et al., 2019) and Numerical Weather Prediction (NWP) models (Coiffier, 2011), have been the cornerstone of weather prediction. These models simulate future weather scenarios by numerically approximating solutions to the differential equations that govern the complex physical dynamics of interconnected atmospheric, terrestrial, and oceanic systems (Nguyen et al., 2023a). Despite significant advancements, these models face notable limitations. Firstly, they are computationally intensive due to the high-dimensional and nonlinear nature of the governing equations (Ren et al., 2021). Secondly, the underlying equations often rely on simplified assumptions about atmospheric dynamics, which can limit their ability to capture intricate, uncommon processes (Palmer et al., 2005). Lastly, these physics-based models typically produce deterministic forecasts based on initial conditions, falling short of explicitly capturing model uncertainties

in weather evolution even though perturbation of initial conditions has been used to represent the input uncertainty (Bülte et al., 2024).

ARIMA (AutoRegressive Integrated Moving Average) is a statistical model widely used for weather prediction (Box et al., 2015). Non-seasonal ARIMA models analyze patterns in historical data but cannot handle seasonality, while seasonal ARIMA extends this framework to account for regular cycles, making it effective for variables like rainfall or temperature (Lai and Dzombak, 2020; Khan et al., 2023). However, ARIMA models have limitations, including difficulty capturing nonlinear relationships, sensitivity to outliers, and the need for careful parameter selection. Bayesian nonparametric nonhomogeneous hidden Markov model is another statistical method that has been studied for predicting daily rainfall (Cao et al., 2024a) and ENSO impacts (Zhang et al., 2024b). However, these methods are usually applied to univariate or low-dimensional responses.

In recent years, data-driven machine learning (ML) and deep learning (DL) models have been increasingly applied to weather and climate modeling, demonstrating remarkable advances in precision, computational efficiency, and uncertainty quantification (Chen et al., 2023d; Nguyen et al., 2023b). For example, deterministic models such as Pangu (Bi et al., 2023) and GraphCast (Lam et al., 2022) have achieved state-of-the-art performance in medium-range (10-day) global weather prediction, surpassing or matching traditional methods in accuracy while dramatically reducing computational costs (up to three orders of magnitude). However, their predictions are often blurry since they are trained by minimizing point-wise loss functions. To overcome this limitation, probabilistic generative models have emerged as powerful tools for weather prediction while achieving uncertainty quantification in those predictions. They consider weather prediction as probabilistic sampling (i.e., generation) conditioning on necessary constraints. Models like CasCast (Gong et al., 2024) and Gencast (Price et al., 2023) leverage probabilistic diffusion techniques for tasks such as precipitation nowcasting and weather prediction, delivering both high-quality predictions and calibrated uncertainty estimates. More recently, foundation models have gained traction in climate and weather modeling as an emerging paradigm (Bodnar et al., 2024; Schmude et al., 2024). These models are pre-trained on massive historical weather

datasets to learn generalizable and comprehensive knowledge, which can then be fine-tuned for diverse downstream tasks (Chen et al., 2023f). Foundation models offer two key advantages: (1) the ability to learn robust and transferable weather representations from large-scale data, and (2) the flexibility to adapt to downstream applications without the need for task-specific models trained from scratch (Miller et al., 2024; Zhu et al., 2024b).

With the rapid advancement of deep learning in weather and climate science, a systematic and up-to-date survey is critical to consolidating knowledge and guiding future research. While several surveys were published in recent years, each has a distinct focus. Ren et al. (2021) reviewed DL models for weather prediction, emphasizing their architectural designs. Molina et al. (2023) summarized DL applications in climate modeling, covering feature detection, extreme weather prediction, downscaling, and bias correction. Moreover, surveys (Fang et al., 2021; Matera et al., 2023) focused on DL techniques for weather forecasting in specific scenarios, such as extreme weather events. Mukkavilli et al. (2023) discussed state-of-the-art DL models across diverse meteorological applications, highlighting their effectiveness over varying spatial and temporal scales. Chen et al. (2023f) categorized DL models for weather and climate science by data modality (e.g., time series, text) and their applications. Distinct from existing surveys, our work provides a novel perspective by reviewing the literature through the lens of training paradigms and offering a broader discussion on future research directions. Our contributions are:

- **Novel Taxonomy.** We introduce a systematic categorization of existing DL models for weather prediction based on their training paradigms: predictive learning, generative learning, and pre-training and fine-tuning.
- **Comprehensive Overview.** We present a detailed survey of the state-of-the-art models, analyzing their strengths, limitations, and applications in weather prediction.
- **Extensive Resources.** We compile an extensive repository of resources, including benchmark datasets, open-source codes, and real-world applications to support further research.
- **Future Directions.** We outline a forward-looking roadmap, highlighting *ten* critical research directions across *five* key avenues to advance DL methods for weather prediction.

2 Background and Preliminaries

2.1 Weather Data Representation

There are two primary types of weather data commonly used: *station-based observation* data and *gridded reanalysis* data. Each offers unique advantages and limitations and both play critical roles in advancing weather and climate research.

Station-Based Observation Data. It originates from weather stations distributed across the globe, collecting high-resolution meteorological measurements at specific locations. These stations provide precise monitoring data, for example, temperature, humidity, wind speed and direction, precipitation, atmospheric pressure, and more. Station-based observations are typically of high temporal resolution, with data recorded hourly or daily, enabling detailed insights into local weather patterns and trends. However, station coverage is often uneven, with a high concentration in populated or economically significant areas and sparse coverage in remote regions such as the oceans, mountains, and deserts. This uneven distribution can limit global-scale analyses, though it remains invaluable for localized forecasting, trend analysis, and model validation.

Gridded Reanalysis Data. It offers a global view by dividing the Earth’s surface into a grid, with each cell assigned values representing averaged weather conditions over its area. It is often called reanalysis data, derived from a combination of sources, including station observations, satellite measurements, and numerical weather prediction (NWP) models. Gridded data provide consistent spatial coverage, including remote areas and oceans, where station-based observations are sparse or nonexistent. Gridded data are typically available at varying resolutions, with common grid sizes ranging from $1^\circ \times 1^\circ$ to $0.25^\circ \times 0.25^\circ$ (each degree corresponds to about 100 km). Temporal resolution can also vary, offering hourly or daily intervals, allowing for detailed temporal analysis.

2.2 Weather Prediction Formulation

As shown in Figure 1, we discuss four types of weather forecasting. (1) *Temporal*: forecasts predict atmospheric variables of interest for future time point(s), $t + \Delta t$, given observation(s) from the recent past. It includes weather and climate forecasts based on the lead time $\Delta t \approx \{\text{hours, days, weeks, months, years}\}$ and encompasses nowcast,

medium-range forecast, sub-seasonal, and seasonal forecast. (2) *Spatial*: methods predict global and regional weather forecasts for any given time point. (3) *Applications*: focus on predicting weather variables of interest. (4) *Event Type*: Weather forecasts may be for extreme events, such as heatwaves, snowstorms, hurricanes, and tropical cyclones. Forecasts could also be for regular, non-extreme periods.

Deterministic weather and climate forecasting can be formulated as follows:

$$[X_{t-(\alpha-1)}, \dots, X_t] \xrightarrow{\mathcal{F}(\theta)} [Y_{t+1}, \dots, Y_{t+\beta}], \quad (1)$$

where X and Y are sets of input and output variables; α and β are the temporal lengths of the input and output windows; $\mathcal{F}(\theta)$ represents the model with the learnable parameters θ . $\mathcal{F}(\cdot)$ can also denote a probabilistic function, i.e., $Y \sim \mathcal{P}(Y|X)$.

2.3 Preliminaries

We identify three types of weather models.

Definition 2.1 (General-Purpose Large Models)

They are typically trained on large, diverse global datasets that include information on multiple meteorological variables of interest, enabling global weather prediction across a broad spectrum of applications.

Definition 2.2 (Domain-Specific Models) *They focus on predicting a single variable, applied to regional weather prediction.*

Definition 2.3 (Foundation Models) *They are large models pre-trained on diverse, massive datasets, allowing for subsequent fine-tuning or adaptation for various downstream tasks.*

Based on the modeling algorithm, we have deterministic and probabilistic training paradigms. Both general-purpose large models and domain-specific models can be trained with deterministic predictive learning (Section 3.1) or probabilistic generative learning (Section 3.2). Foundation Models are pre-trained and then fine-tuned (Section 3.3).

3 Overview and Taxonomy

This section provides an overview and categorization of deep learning (DL) models for weather forecasts. Our survey mainly focuses on three aspects: modeling paradigm, model backbone, and application domain. The modeling paradigm includes deterministic *predictive learning*, probabilistic *generative learning*, and *pre-training and fine-tuning*

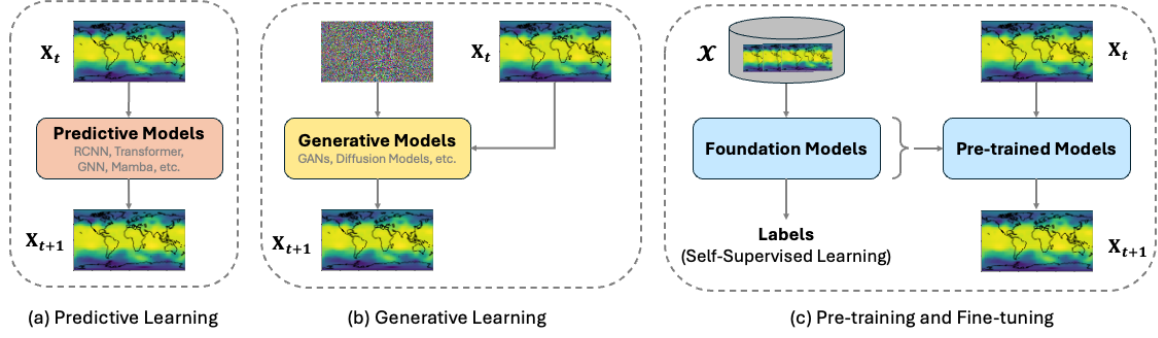


Figure 2: The illustration of various frameworks of training deep learning models on weather prediction. For clarity, this visualization focuses exclusively on single-step predictions for a single variable.

(see Figure 2). Weather and climate models can be categorized based on model backbones, such as Recurrent Neural Networks, Transformers, Graph Neural Networks, Mamba, Generative Adversarial Networks, and Diffusion Models. The theoretical details of these models are provided in Appendix B. At the application level, the existing models can be divided into general-purpose and domain-specific models. We present a detailed comparison and summary in Table 1 and Figure 3.

Table 1: General-Purpose Large Models vs Domain-Specific Models.

	General-Purpose Large Models	Domain-Specific Models
Scope	Global, multi-variable	Regional forecasts, single-variable
Spatial	Coarse ($0.25^\circ \sim 5.625^\circ$)	High ($\leq 0.1^\circ$)
Temporal	Coarse (6 12 hours)	High (5 mins $\sim$ 1 hour)
Training Data	≥ 10 Years	Days, Months, Years
Architectures	Transformer, GNN	Transformer, GNN, RNN, CNN, Mamba

3.1 Predictive Learning

Predictive learning methods are usually *deterministic*, where models aim to predict future states of weather variables (like temperature, humidity, wind speed, and precipitation) based on past and present observations. These models are typically built to recognize weather patterns or dependencies in historical data by minimizing a point-wised loss function (e.g., mean absolute errors). We systematically categorize these predictive models into general-purpose large models and domain-specific models. Each categorization is discussed with various model architectures.

3.1.1 General-Purpose Large Models

Large Language Models (LLMs) (Zhao et al., 2023) have garnered significant attention in recent years. Similarly, large-scale weather models have been developed to address global weather prediction tasks across multiple meteorological variables, leverag-

ing deterministic predictive frameworks.

Transformer-based models. Transformer models (Vaswani, 2017) are widely used as a backbone. FourCastNet (Pathak et al., 2022) is developed for global data-driven weather forecasting by employing a Fourier transform-based token-mixing scheme (Guibas et al., 2021) with a vision transformer (ViT) (Dosovitskiy et al., 2020). The multiple-time step prediction is achieved by using trained models in autoregressive inference mode. FengWu (Chen et al., 2023a) processes each weather variable separately, using multiple encoders to extract individual feature embeddings. Then, an elaborately designed transformer network fuses these embeddings to capture the interaction among all variables. As with FourCastNet, Fengwu also autoregressively forecasts multiple steps over a long range. FengWu-4DVar (Xiao et al., 2023) integrates FengWu with the Four-Dimensional Variational (4DVar) assimilation algorithm (Rabier et al., 1998), accomplishing both global weather forecasting and data assimilation. SwinVRNN (Hu et al., 2023) utilizes the Swin Transformer (Liu et al., 2022) and RNN for weather prediction, but with a perturbation module to generate ensemble forecasts. SwinRDM (Chen et al., 2023b) uses Swin-RNN for prediction and a diffusion model for super-resolution output. HEAL-ViT (Ramavajjala, 2024) explores Vision Transformers on a spherical mesh, benefiting from both spatial homogeneity inherent in graphical models and efficient attention mechanisms. The TianXing model (Yuan et al., 2025) proposes a variant attention mechanism with linear complexity for global weather prediction, significantly diminishing GPU resource demands, with only a marginal compromise in accuracy.

While these models have achieved impressive performance, any iterative inference process ac-

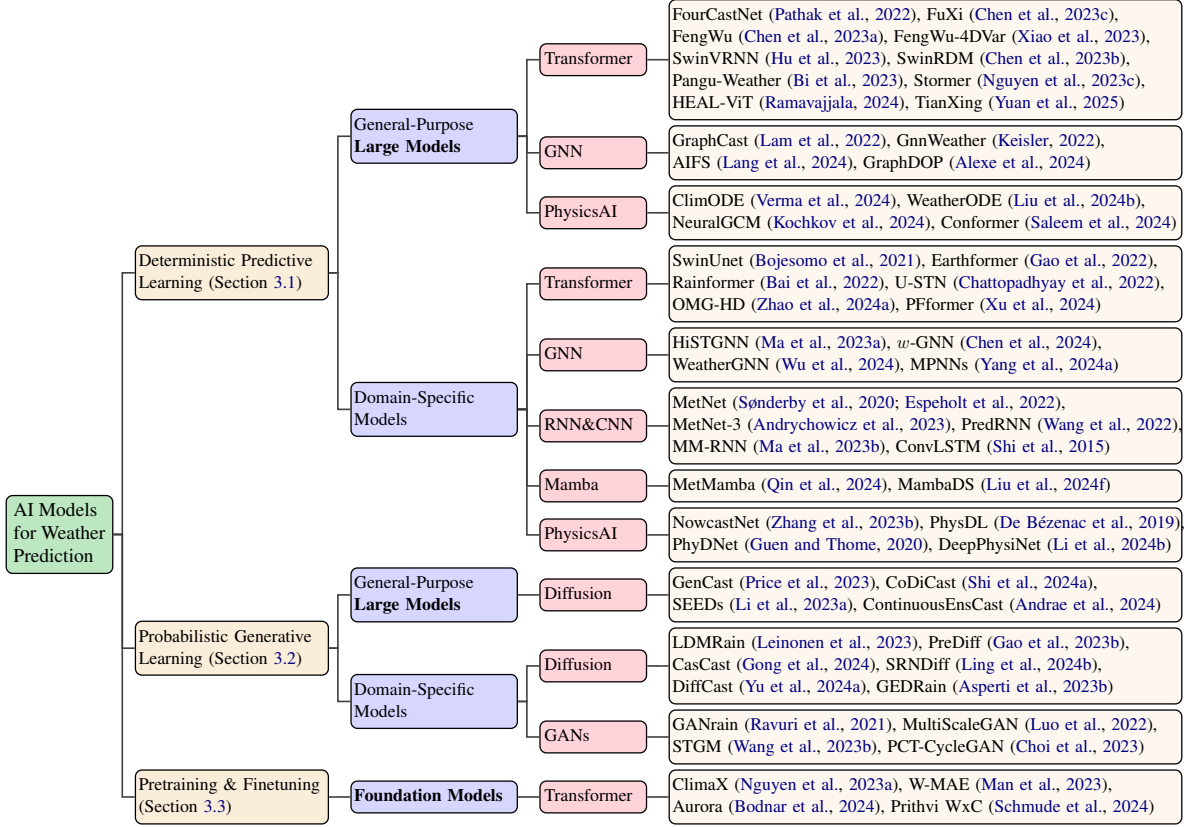


Figure 3: A comprehensive taxonomy of deep learning and foundation models for weather prediction from the perspectives of training paradigms (dark yellow), model scopes (purple), and model architectures (pink).

cumulates errors as the length of the prediction window increases. The Pangu-Weather (Bi et al., 2023) model uses a hierarchical temporal aggregation algorithm to alleviate cumulative forecast errors. They train four individual models for lead times of 1, 3, 6, and 24 hours. In the testing stage, the greedy algorithm is used to guarantee the minimal number of iterations of the trained models for a forecast window. Furthermore, they design a 3D Earth Specific Transformer (3DEST) architecture that formulates the height (pressure level) information into cubic data, capturing more intricate spatiotemporal dynamics. Similarly, the FuXi model (Chen et al., 2023c) employed a combination of FuXi-Short, FuXi-Medium, and FuXi-Long models to produce 15-day forecasts, where each model generates 5-day forecasts. Its backbone is a U-transformer, coupling U-Net (Ronneberger et al., 2015), and a Swin Transformer (Liu et al., 2022). In addition to the integration of direct and iterative forecasting, the Stormer model (Nguyen et al., 2023c) needs the explicit time point, $t + \Delta t$ to guide the models for predictions.

GNN-based models. Keisler (2022) introduced an approach to global weather prediction using

graph neural networks (GNNs) (Wu et al., 2020). By modeling the Earth as a graph with nodes representing spatial locations and edges encoding their relationships, the model captures spatial dependencies in weather patterns. This GNN-based method effectively integrates local and global weather dynamics. Another GNN-based model, GraphCast (Lam et al., 2022), forecasts hundreds of weather variables with a longer forecast range (up to 10 days ahead) at a higher spatial resolution (0.25 degree) after training with reanalysis gridded ERA5 data (Rasp et al., 2023). It also provides better support for severe weather compared to the European Centre for Medium-Range Weather Forecasts (ECMWF)’s High-RESolution forecast (HRES), a component of the Integrated Forecast System (IFS). More recently, ECMWF also proposed GNN-based models, AIFS (Lang et al., 2024) and GraphDOP (Alexe et al., 2024). The latter is a model that operates solely on inputs and outputs in observation space, with no gridded climatology and/or NWP (re)analysis inputs or feedback.

Physics-AI-based models. Although data-driven methods have demonstrated high accuracy and efficiency, they operate as black-box models that fre-

quently overlook underlying physical mechanisms, such as turbulence, convection, and atmospheric airflow. ClimODE (Verma et al., 2024) implements a key principle of *advection* to model a spatiotemporal continuous-time process, namely, weather changes due to the spatial movement over time. It aims to precisely describe the value-conserving dynamics of weather evolution with continuity ODE (Marchuk, 2012), learning global weather transport as a neural flow. It also includes a Gaussian emission network for predicting uncertainties and source variations. To solve the advection equation more accurately, WeatherODE (Liu et al., 2024b) adopts wave equation theory (Evans, 2022) and a time-dependent source model and designs the CNN-ViT-CNN sandwich structure, facilitating efficient learning dynamics tailored for distinct yet interrelated tasks with varying optimization biases. NeuralGCM (Kochkov et al., 2024) employs a differentiable dynamical core for solving *more* primitive equations, including momentum equations, the second law of thermodynamics, a thermodynamic equation of state, continuity equation, and hydrostatic approximation. It also develops a learned physics module that parameterizes physical processes with a neural network, predicting the effect of unresolved processes such as cloud formation, radiative transport, precipitation, and subgrid-scale dynamics. Conformer (Saleem et al., 2024) is a spatiotemporal Continuous Vision Transformer for weather forecasting, learning the continuous weather evolution over time by implementing continuity in the multi-head attention mechanism.

3.1.2 Domain-Specific Models

We present domain-specific predictive models for regional or single-variable weather predictions.

Transformer-based models. SwinUnet (Bojesomo et al., 2021) employs the hybrid model of Swin Transformer and U-Net for regional weather forecasts in Europe. Earthformer (Gao et al., 2022) proposes a generic, flexible, and efficient space-time attention block (Cuboid Attention) Earth system forecasting, which can decompose the data into cuboids and apply cuboid-level self-attention in parallel. Rainformer (Bai et al., 2022) combines CNN and Swin Transformer for precipitation nowcasting. PFformer (Xu et al., 2024) utilizes i-Transformer (Liu et al., 2023a) to learn spatial dependencies among multiple observation stations for short-term precipita-

tion forecasting. Vision transformer (Dosovitskiy et al., 2020) has been applied to estimate lightning intensity in Ningbo City, China (Lu et al., 2022). NowcastingGPT (Meo et al., 2024) develops Transformer-based models with Extreme Value Loss (EVL) regularization (von Bortkiewicz, 1921) for extreme precipitation nowcasting. The U-STN model (Chattopadhyay et al., 2022) integrates data assimilation with a deep spatial-transformer-based U-NET to predict the global geopotential while the OMG-HD model (Zhao et al., 2024a) leverages the Swim Transformer for regional high-resolution weather forecast trained with multiple observational data, including stations, radar, and satellite.

GNN-based models. HiSTGNN (Ma et al., 2023a) incorporates an adaptive graph learning module comprising a global graph representing regions and a local graph capturing meteorological variables for each region. The w -GNN model (Chen et al., 2024) leverages Graph Neural Networks coupled with physical factors for precipitation forecast in China. WeatherGNN (Wu et al., 2024) proposes a fast hierarchical Graph Neural Network (FHGNN) to extract the spatial dependencies. The MPNN model (Yang et al., 2024a) exploits heterogeneous GNNs for both station-observed and gridded weather data, where the node at the prediction location aggregates information from its heterogeneous neighboring nodes by message passing.

RNN- & CNN-based models. The ConvLSTM model (Shi et al., 2015) couples CNNs and LSTMs as the model backbone for precipitation nowcasting, usually with a lead time between 1 to 3 hours. Similar works include MetNet-1 (Sønderby et al., 2020) and MetNet-2 models (Espeholt et al., 2022) for precipitation forecasting for lead times of 8 and 12 hours. MetNet-3 (Andrychowicz et al., 2023) significantly extends both the lead times (up to 24 hours) and variables (precipitation, wind, temperature, and dew point) by learning from both dense and sparse data sensors. MM-RNN (Ma et al., 2023b) introduces knowledge of elements to guide precipitation prediction and learn the underlying atmospheric motion laws using RNNs. Based on the original LSTMs, PredRNN (Wang et al., 2022) proposes a zigzag memory flow that propagates in both a bottom-up and top-down fashion across all layers, enabling the dynamic communication at various levels of RNNs. Other variants of ConvLSTM for precipitation nowcasting include TrajGRU (Shi et al., 2017) and Predrnn++ (Wang et al., 2018).

Mamba-based models. MetMamba (Qin et al., 2024) exploits Mamba’s selective scan to achieve token (spatial, temporal) mixing and channel mixing to capture more complex spatiotemporal dependencies in weather data. MambaDS (Liu et al., 2024f) attempts to use the selective state space model (Mamba) for the meteorological field downscaling. VMRNN (Tang et al., 2024) develops an innovative architecture tailored for spatiotemporal forecasting by integrating Vision Mamba and LSTM, surpassing established vision models in both efficiency and accuracy.

Physics-AI-based models. NowcastNet (Zhang et al., 2023b) is a nonlinear nowcasting model for extreme precipitation that unifies physical-evolution schemes and conditional-learning methods into a neural network framework. PhysicsAI (Das et al., 2024) has evaluated NowcastNet model with a case study on the Tennessee Valley Authority (TVA) service area, outperforming the High Resolution Rapid Refresh (HRRR) model. PhysDL (De Bézenac et al., 2019) presents how physical knowledge (*advection* and *diffusion*) could be used as a guideline for designing efficient deep-learning models, exemplifying sea surface temperature predictions. PhyDNet (Guen and Thome, 2020) is a two-branch deep learning architecture that explicitly disentangles known PDE dynamics from unknown complementary information. DeepPhysiNet (Li et al., 2024b) incorporates atmospheric physics into the loss function of deep learning methods as hard constraints for accurate weather modeling.

More generally, we provide state-of-the-art predictive models for time series forecasting across various domains. While these models are not specific for weather modeling, they offer insightful modeling advancements since weather data is often represented as time series. Representative models include but not limited to iTransformer (Liu et al., 2023a), PatchTST (Nie et al., 2022), FEDformer (Zhou et al., 2022), DLinear (Zeng et al., 2023), Autoformer (Chen et al., 2021a). More recently, Han et al. (2024b) collected worldwide meteorological monitoring data, created a benchmark dataset, and completed a comprehensive evaluation with those advanced models above.

3.2 Generative Models

Generative models can be used for weather *prediction* by treating them as *generative* processes condi-

tioned on observations from the past. More significantly, since these generative models are probabilistic, they are well suited to generate ensemble forecasts that can help quantify the uncertainty in the predictions, facilitating informed decision-making.

3.2.1 General-Purpose Large Models

Diffusion-based models. Some researchers have developed generative models for global weather prediction. GenCast (Price et al., 2023) uses diffusion models for probabilistic weather forecasts conditioning on the past two observations, generating an ensemble of stochastic 15-day global forecasts, at 12-hour steps and 0.25° latitude-longitude resolution, for over 80 surface and atmospheric variables. As a variant of GenCast, CoDiCast (Shi et al., 2024a) leverages a *pre-trained* encoder to learn embeddings from observations from the recent past and a *cross-attention* mechanism to guide the generation process to predict future weather states. Similar work includes SEEDs (Li et al., 2023a) for the global weather forecast. The three methods above are trained on a single forecasting step and rolled out autoregressively. However, they are computationally expensive and accumulate errors for high temporal resolution due to the many rollout steps. ContinuousEnsCast (Andrae et al., 2024) addresses these limitations by proposing a continuous forecasting diffusion model that takes lead time as input and forecasts the future weather state in a single step while maintaining a temporally consistent trajectory for each ensemble member.

3.2.2 Domain-Specific Models

Here we discuss domain-specific models for generative learning with generative adversarial networks (GANs) (Goodfellow et al., 2014; Mirza, 2014) and diffusion models (Ho et al., 2020).

GAN-based models. GANrain (Ravuri et al., 2021) employs a conditional generative adversarial network (GAN) for the precipitation prediction problem, where the generator generates future precipitation frames and the discriminator learns to distinguish whether a sample is coming from the original training data or was generated by the generator. MultiScaleGAN (Luo et al., 2022) evaluates GANs (Goodfellow et al., 2014) and WassersteinGAN (Arjovsky et al., 2017) for precipitation nowcasting in Guangdong province, China, and indicates that GAN-based models outperform the traditional ConvGRU, ConvLSTM, and multiscale

CNN models. STGM (Wang et al., 2023b) introduces a task-segmented, synthetic-data generative model (STGM) for heavy rainfall nowcasting by utilizing real-time radar observations in conjunction with physical parameters derived from the Weather Research and Forecasting (WRF) model. PCT-CycleGAN (Choi et al., 2023) extends the idea of the cycle-consistent adversarial networks (CycleGAN) (Zhu et al., 2017) and proposes a paired complementary temporal CycleGAN for radar-based precipitation nowcasting.

Diffusion-based models. LDMRain (Leinonen et al., 2023) uses the architecture of latent diffusion model (Rombach et al., 2022) for precipitation nowcasting – short-term forecasting based on the latest observational data. Similar works include SRNDif (Ling et al., 2024b) and GEDRain (Asperti et al., 2023b). DiffCast (Yu et al., 2024a) models the precipitation process from two perspectives: the deterministic component accounts for predicting a global motion trend by a coarse forecast, while the stochastic component aims to learn local stochastic variations with the residual mechanism. CasCast (Gong et al., 2024) develops a cascaded framework consisting of a deterministic predictive model to output blurry predictions, and a probabilistic diffusion model with inputs as both past observations and deterministic predictions beforehand. Because the deterministic predictions are the future frames, such frame-wise guidance in the diffusion model can provide a frame-to-frame correspondence between blurry predictions and latent vectors, resulting in a better generation of small-scale patterns. However, directly applying diffusion models might generate physically implausible predictions. To tackle these limitations, Prediff (Gao et al., 2023b) proposes a conditional latent diffusion model for probabilistic forecasts and then aligns forecasts with domain-specific physical constraints. This is achieved by estimating the deviation from imposed constraints at each denoising step and adjusting the transition distribution accordingly.

TimeDiff (Shen and Kwok, 2023), TimeDDPM (Dai et al., 2023), LTD (Feng et al., 2024b), TimeGrad (Rasul et al., 2021), and Dyffusion (Rühling Cachay et al., 2024) are examples that have applied diffusion models to general time series modeling, which could be adapted to weather time series. Yang et al. (2024b) provides a comprehensive survey of such methods.

3.3 Foundation Models

Foundation Models (FMs) have garnered significant research interest due to their powerful prior knowledge acquired through pre-training on massive data and their remarkable adaptability to downstream tasks with fine-tuning strategies (He et al., 2024c). While foundation models may be large language models (LLMs), a few foundation models in the weather domain have been proposed.

ClimaX (Nguyen et al., 2023a) is a versatile and generalizable deep-learning model developed for weather and climate science. It is trained on heterogeneous datasets encompassing diverse variables, spatiotemporal coverage, and physical principles with CMIP6 datasets and it can be fine-tuned for a wide range of weather and climate applications, including those involving atmospheric variables and spatiotemporal scales not encountered during pre-training. W-MAE (Man et al., 2023) is pre-trained with similar data, but using reconstruction tasks with the Masked Autoencoder model (He et al., 2022). The pre-trained model can be fine-tuned for various tasks, e.g., multi-variate forecasting. Aurora (Bodnar et al., 2024) is a large-scale foundation model pre-trained on over a million hours of diverse weather and climate data. Unlike the two foundation models above, Aurora can be fine-tuned in one of two ways: short-time fine-tuning (i.e., fine-tuning the entire architecture through one or two roll-out steps) and rollout fine-tuning for long-term multi-step predictions with low-rank adaption (LoRA) (Hu et al., 2021a). Prithvi WxC (Schmude et al., 2024) is a foundation model with 2.3 billion parameters developed using 160 variables. It is essentially a scalable and flexible 2D vision transformer with varying sizes of tokens or windows. During the pre-training, the Masked Autoencoder model (He et al., 2022) is pre-trained by masking different ratios of tokens and windows to capture both regional and global dependencies in the input data. It can be fine-tuned for nowcasting, forecasting, and downscaling tasks. More recently, AtmosArena (Nguyen et al., 2024) benchmarks foundation models for atmospheric sciences across various atmospheric variables.

The large foundation models designed for general time series data, including TimeFM (Das et al., 2023), Moment (Goswami et al., 2024), Timer (Liu et al., 2024d), Moirai (Woo et al., 2024), and Chronos (Ansari et al., 2024) may be adapted for weather forecasting.

4 Applications and Resources

This section introduces the diverse applications of deep learning models in weather and climate science. We provide an overview of the available datasets, summarized in detail in Table 3 in Appendix A.

4.1 Precipitation

Precipitation prediction has witnessed significant advances driven by deep learning (DL) applications, focusing mainly on precipitation nowcasting (Gao et al., 2020, 2021; Ashok and Pekkat, 2022; Verma et al., 2023; Salcedo-Sanz et al., 2024; An et al., 2024). CNN-based architectures, particularly U-Net, have been widely utilized for their ability to extract local features through convolutional layers, effectively capturing high-dimensional spatiotemporal dynamics of precipitation (Lebedev et al., 2019; Ayzel et al., 2020b; Han et al., 2021; Ehsani et al., 2022; Seo et al., 2022; Kim et al., 2022a; Zhang et al., 2023b). RNN-based models, Transformers, and their hybrid designs combining convolutions represent another dominant approach, optimized for long-term dependency modeling (Shi et al., 2015; Wang et al., 2017; Park et al., 2022; Gao et al., 2022; Bai et al., 2022; Geng et al., 2024; Bodnar et al., 2024; Zhao et al., 2024b; Schmude et al., 2024). Generative models have also played a critical role, with adversarial models (e.g., GANs) (Jing et al., 2019; Liu and Lee, 2020; Ravuri et al., 2021; Wang et al., 2023c; She et al., 2023; Choi et al., 2023; Yin et al., 2024; Franch et al., 2024) contributing to precipitation synthesis. Moreover, probabilistic generative diffusion models have gained attention for their superior stability, controllability, and fine-grained synthesis capabilities (Leinonen et al., 2023; Gao et al., 2023b; Yu et al., 2024a; Gong et al., 2024).

4.2 Air Quality

Air quality prediction is of critical importance to society. Zheng et al. (2013) employ artificial neural network (ANN) with spatially-related features to predict the air quality in Beijing. Waseem et al. (2022) employed a CNN-Bi-LSTM architecture for air quality prediction in Xi'an, China, and Yi et al. (2018) propose a model combining a spatial transformation component and a deep distributed fusion network to predict air quality in nine major cities in China. More recently, Shi et al. (2022) evaluate various deep learning models, including

RNNs, LSTMs, GRUs, and Transformers, for air quality prediction in Beijing. Nationwide air quality forecasting in China has leveraged advanced architectures such as hierarchical group-aware graph neural networks (GAGNN) (Chen et al., 2023e), spatiotemporal graph neural networks (STGNNs) (Wang et al., 2020), and Transformer-based models (Liang et al., 2023; Yu et al., 2025). Additionally, RNNs have been utilized for air quality prediction in India (Arora et al., 2022) and Pakistan (Waseem et al., 2022), while hybrid CNN-LSTM architectures have been applied for predictions in Barcelona and Turkey (Gilik et al., 2022).

4.3 Sea Surface Temperature

The change in Sea Surface Temperature can cause El Niño/Southern Oscillation (ENSO) and La Niña phenomena, largely impacting the global extreme climate, such as increasing the chances of floods, droughts, heat waves, and cold seasons (Wang et al., 2023a). Niño 3.4 index, an important indicator for ENSO prediction, has been predicted using different deep learning (DL) models, such as RNN-based (Huang et al., 2019; Geng and Wang, 2021), CNN-based (Ham et al., 2019; Liu et al., 2021), residual CNNs (Hu et al., 2021b), ConvLSTM (He et al., 2019), GNN-based (Cachay et al., 2020), and Transformer-based models (Ye et al., 2021; Zhou and Zhang, 2023; Song et al., 2023). More recently, an adaptive graph spatial-temporal attention network (AGSTAN) has been proposed for longer lead (i.e., 23 months) ENSO prediction (Liang et al., 2024). Mu et al. evaluates multiple DL models for the Niño 3 index, Niño 3.4 index, and Niño 4 index with a multivariate air-sea coupler. Similar evaluation work involves comparing deep learning models for ENSO forecasting and presenting ENSO dataset (Mir et al., 2024). Moreover, some researchers directly predict the sea surface temperature using spatiotemporal graph attention networks (Gao et al., 2023c) and physical knowledge-enhanced generative adversarial networks (Meng et al., 2023). ENSO impacts have also been studied, including river flows (Liu et al., 2023b), rainfall (He et al., 2024b), and heat-waves (He et al., 2024a).

4.4 Flood

Accurate flood prediction is essential for mitigating the adverse impacts of flooding. Recent advances in deep learning (DL) have led to the development of various models tailored for flood forecasting

and mapping, such as CNN-based (Adikari et al., 2021), RNN-based and LSTM (Nevo et al., 2022; Ruma et al., 2023), and CNN-RNN hybrid models such as ConvLSTM (Li et al., 2022; Moishin et al., 2021), and LSTM-DeepLabv3+ (Situ et al., 2024a). Situ et al. (2024b) employs the *attention* mechanism for urban flood damage and risk assessment with improved flood prediction and land use segmentation. Furthermore, graph-based models have also gained attention for flood prediction (Kirschstein and Sun, 2024). FloodGNN-GRU combines GNNs and Gated Recurrent Units (GRUs) for spatiotemporal flood prediction by incorporating vector features like velocities (Kazadi et al., 2024) while Graph Transformer Network (FloodGTN) integrates GNNs and Transformers to learn spatiotemporal dependencies in water levels (Shi et al., 2023, 2024b). Additionally, physics-guided models further enhance flood prediction by embedding physical laws into model training. For instance, the DK-Diffusion model incorporates flood physics into its loss function to align predictions with hydrological principles (Shao et al., 2024). DRUM leverages diffusion model for operational flood forecasting and long-term risk assessment (Ou et al., 2024). Moreover, conditional GANs have been explored for flood predictions across untrained catchments (do Lago et al., 2023), demonstrating their versatility in diverse hydrological conditions.

4.5 Drought

Drought, driven by a complex interplay of meteorological, agricultural, hydrological, and socioeconomic factors, manifests across diverse spatial and temporal scales (Wilhite, 2016; Gyaneshwar et al., 2023). We focus on DL methods that consider meteorological drivers, such as precipitation deficits, wind patterns, and temperature anomalies, to predict various drought indices. LSTMs have been widely used to predict spatial precipitation patterns (dry-wet) (Gibson et al., 2021) and drought indices related to precipitation, such as the standardized precipitation index (SPI) (Poornima and Pushpalatha, 2019; Dikshit and Pradhan, 2021) and the standardized precipitation evapotranspiration index (SPEI) (Tian et al., 2021; Dikshit et al., 2021; Xu et al., 2022), excelling at capturing long-term dependencies. Beyond SPI and SPEI (Adikari et al., 2021; Dhyani and Pandya, 2021; Hao et al., 2023), CNNs have been applied for predicting other indices, such as the soil moisture index

(SMI) (Dhyani and Pandya, 2021) and soil moisture condition index (SMCI) (Zhang et al., 2024c), aiding agricultural drought prediction. Hybrid models like ConvLSTM and CNN-LSTM have demonstrated significant improvements in multi-temporal predictions for SPEI (Danandeh Mehr et al., 2023; Nyamane et al., 2024) and SPI (Park et al., 2020), as well as indices like the scaled drought condition index (SDCI) (Park et al., 2020), composite drought index (CDI) (Zhang et al., 2023a), and Palmer drought severity index (PDSI) (Elbeltagi et al., 2024). Specifically, the CNN-GRU model has effectively forecasted daily reference evapotranspiration (ET) (Ahmed et al., 2022). Swin Transformer was used for drought prediction across multiple scales (Zhang et al., 2024a). Meanwhile, GANs have emerged as robust tools for drought prediction, with applications spanning vegetative drought prediction (Shukla and Pandya, 2023), and SMI (Ferchichi et al., 2024).

4.6 Tropical Storms/Cyclones and Hurricanes

Accurate forecasting of tropical storms, cyclones, and hurricanes is crucial for mitigating their devastating impacts. CNN-based models have been increasingly employed to predict various aspects of these phenomena, focusing on targets such as storm formation (Zhang et al., 2021; Nguyen and Kieu, 2024), intensity (Kim et al., 2024), track (Giffard-Roisin et al., 2020; Lian et al., 2020), and associated rainfall (Kim et al., 2022b). Hybrid models, such as CNN-LSTM, further improve the accuracy of intensity prediction (Alijoyo et al., 2024), extend lead times up to 60 hours (Kumar et al., 2022), and effectively capture landfall in terms of location and time (Kumar et al., 2021). GANs have also proven valuable in downscaling tropical cyclone rainfall to hazard-relevant spatial scales (Vosper et al., 2023) and in multitask frameworks for simultaneously forecasting cyclone paths and intensities (Wu et al., 2021). Recent approaches like diffusion models have been explored for forecasting cyclone trajectories and precipitation patterns (Nath et al., 2023). GNNs integrated with GRUs have been utilized to model storm surge dependencies across observation stations, offering improvements in spatial and temporal forecasting (Jiang et al., 2024).

4.7 Wildfire

Accurate wildfire prediction is critical for disaster management and mitigation. CNN-based models have demonstrated strong capabilities in wildfire

spread prediction (Khennou et al., 2021; Shadrin et al., 2024), including forecasting fire weather with high spatial resolution (Son et al., 2022), generating spread maps (Huot et al., 2022), and modeling large-scale fire dynamics using multi-kernel architectures (Marjani and Mesgari, 2023). RNNs, including GRUs and LSTMs, excel in modeling wild-fire risk and predicting spread, with GRU-LSTM showing superior performance in longer time series data (Perumal and Van Zyl, 2020; Dzulhijjah et al., 2023; Gopu et al., 2023). Hybrid CNN-LSTM models further enhance prediction accuracy, offering near-real-time daily wildfire spread forecasting (Marjani et al., 2024) and incorporating multi-temporal dynamics for prediction (Marjani et al., 2023). ConvLSTM models capture a wide range of temporal scales in wildfire prediction, from short-term intervals of 15 minutes (Burge et al., 2023) to longer-term forecasts extending up to 10 days (Masrur and Yu, 2023; Masrur et al., 2024). Other advancements include GANs, which have been utilized for wildfire risk prediction through conditional tabular data augmentation (Chowdhury et al., 2021), and GNNs, which simulate wildfire spread in variable-scale landscapes, effectively addressing landscape heterogeneity (Jiang et al., 2022). Additionally, researchers have also explored Transformer models for wildfire prediction (Miao et al., 2023; Cao et al., 2024b).

5 Challenges and Future Directions

In this section, we introduce primary challenges and suggest promising future research opportunities from the perspectives of DL models (Subsections 5.1-5.4) and data (Subsections 5.4-5.5).

5.1 Trustworthy AI

We discuss trustworthy AI models paying careful attention to robustness, generalization, explainability, scalability, and uncertainty quantification.

Robustness: Weather data is often subject to observational or collection biases, leading to significant performance degradation in AI models. These biases may stem from inconsistent data collection methods, non-uniformity or limited spatial or temporal coverage, and inaccuracies in sensor measurements. As a result, AI models trained on such biased data sets may struggle to generalize effectively. **Opportunities:** (1) Bias correction with statistical adjustments (Durai and Bhadrwaj, 2014) and data assimilation (Berry and Harlim, 2017) can

be applied to reduce biases in the data. (2) Adversarial training (Wang et al., 2024), a technique originally developed to defend against adversarial attacks in machine learning, can mitigate vulnerabilities by exposing models to challenging or perturbed examples during training, allowing them to generalize better to real-world biases or anomalies. It involves creating perturbed versions of weather data representing scenarios with systematic biases and incorporating adversarial examples alongside clean data during training to improve its robustness to biased data sets (Schmalfuss et al., 2023).

Generalization: AI models often fail to perform effectively on rare extreme weather or anomalous events that fall outside the distribution (OOD) of the training samples. **Opportunities:** (1) Physical laws represent precious wisdom from domain pioneers, but they are rarely explicitly incorporated into AI models (Feng et al., 2023). Leveraging physics-informed or physics-guided AI approaches can increase reliability and consistency with the physical world (Chen et al., 2021b; Meng et al., 2021; Yin et al., 2023), particularly while addressing extreme or unseen scenarios. Although significant progress has been made in the integration of physics and AI (see “Physics-AI” in Section 3), further exploration is needed to optimize and refine these approaches. (2) DL models perform poorly in extreme weather events due to their rarity and limited representation in the training data. Effective data augmentation with generative diffusion models (Trabucco et al., 2023; Mardani et al., 2023) is a promising method to address or alleviate this challenge. By augmenting the training set with more extreme samples, DL models are better equipped to understand these rare events comprehensively, enhancing their generalizability. Therefore, it is worth exploring how to effectively augment data with extreme samples.

Explainability: Neural networks are frequently referred to as “black boxes” due to the opacity of their internal processes, making it challenging to interpret how they produce outputs (Guidotti et al., 2018). In the weather and climate domains, understanding the underlying mechanisms of these models is of paramount importance and a necessity to ensure reliability and trustworthiness. **Opportunities:** Explainable AI tools, such as SHAP (Shapley Additive Explanations) (Lundberg, 2017), LIME (Local Interpretable Model-Agnostic Explanations) (Ribeiro et al., 2016), Grad-

CAM (Selvaraju et al., 2017), and causal analysis (Zhang et al., 2011) have gained prominence in addressing this challenge. Furthermore, the principle of information bottleneck (IB) has been used for explainable learning in the time series domain (Feng et al., 2024a; Liu et al., 2024e). Given that weather data inherently constitute time series, we advocate exploring how the information bottleneck method can enhance the explainability of weather modeling. Leveraging these techniques can help determine whether DL models are producing meaningful results based on legitimate patterns or merely fabricating outputs, reinforcing trustworthiness and accountability in model predictions.

Varying Resolution: In weather and climate science, it is common to deal with varying data resolutions. For example, weather data have differing temporal and spatial resolutions across modalities. Meteorological observations might have an hourly temporal resolution from sparse sensors, radar echo data could feature six-minute temporal intervals and a spatial resolution of 1–4 km, and satellite imagery might exhibit a temporal resolution of 30 minutes with a spatial resolution of 5–12 km. These discrepancies complicate the task of harmonizing information across modalities for robust model development (Chen et al., 2023f). **Opportunities:** Therefore, an important challenge is to build models that can handle training data of varying resolutions and also reliably predict at a different resolution. Such models could revolutionize how we integrate data from various sources, including observations, satellite imagery, and numerical simulations, which often differ in granularity and format. Aurora processes input data with varying patch sizes (Bodnar et al., 2024), and IPOT (Inducing-point operator transformer) uses a smaller number of inducing points, flexibly handling any discretization formats of input (Lee and Oh, 2024).

Uncertainty Quantification: Given the chaotic nature of the atmosphere, quantifying uncertainty in weather predictions is essential to allow informed decision-making. Approaches such as initial conditions perturbation and Monte Carlo dropout have been studied (Bülte et al., 2024); however, they only simulate the aleatoric uncertainty, i.e., the inherent randomness in from weather data or the epistemic uncertainty from the model itself due to the limited knowledge. **Opportunities:** Generative diffusion models address both aleatoric and epistemic uncertainty simultaneously. Diffu-

sion models learn the full probability distribution of the data, capturing aleatoric uncertainty through stochastic sampling, where the spread of outcomes reflects inherent data variability. When conditioned on the inputs, added stochastic noise incorporates input variability, further representing data-driven uncertainty. Additionally, by initializing from different noise points, diffusion models capture epistemic uncertainty (Du and Li, 2023; Price et al., 2023), with greater variability in regions of sparse training data. This inherent stochasticity makes diffusion models a robust tool for quantifying both aleatoric and epistemic uncertainties.

5.2 Retrieval-augmented Foundation Models

Retrieval-augmented generation (RAG) (Gao et al., 2023a) has emerged as a promising approach to enhance foundation models by integrating external domain knowledge. **Opportunities:** While RAG has been extensively explored in domains such as medicine (Xiong et al., 2024), its application to weather and climate modeling remains underexplored. Depending on whether the foundation model uses diffusion models (Yang et al., 2023) or large language models (LLMs) (Zhao et al., 2023) as its underlying architecture, different opportunities arise for leveraging retrieval augmentation: (1) Diffusion Models for Weather Forecasting: In the context of diffusion-based weather models (Shi et al., 2024a), retrieval augmentation can be leveraged to fetch historical weather patterns similar to the current state, allowing it to recreate historical conditions that may have appeared in the past and that can serve as references to refine predictions, potentially improving accuracy and robustness (Liu et al., 2024a; Ravuru et al., 2024). It holds significant potential to enhance performance in extreme weather scenarios by addressing the challenges posed by data rarity. (2) LLMs for Weather Text Analysis: For tasks involving textual analysis of weather-related corpora, such as extreme weather reports or climatological summaries (Colverd et al., 2023), retrieval augmentation can provide valuable context by identifying and incorporating relevant documents. This approach can significantly enhance the model’s ability to generate informed and contextually relevant outputs (Juhász et al., 2024). By bridging retrieval-based methodologies with foundation models, RAG helps to maximize the power of foundation models, presenting an exciting avenue for advancing both accuracy and interpretability in weather and climate applications.

5.3 Generative AI with Weather Constraints

Generative models have achieved enormous success in image generation (Goodfellow et al., 2014; Ho et al., 2020). More interestingly, controllable generative models can synthesize customized images according to conditions provided by users (Gauthier, 2014; Rombach et al., 2022).

Opportunities: In the weather domain, weather *prediction* can be formulated as weather *generation* conditioned on temporal and spatial similarities. These conditions or constraints could come from (1) partial differential continuity equations (Broomé and Ridenour, 2014; Palmer, 2019), which describe the weather as a flux, a spatial movement of quantities over time; (2) Tobler’s First law of Geography (Tobler, 2004), which states that everything is related to everything else, but near things are more related than distant things; and (3) Tobler’s Second law of Geography (Tobler, 1999), which states that the phenomenon external to a geographic area of interest affects what goes on inside; and (4) other modalities, such as station-based, satellite-based (Qu et al., 2024; Xiang et al., 2024), and even text data (Li et al., 2024a). By leveraging the weather constraints as prior knowledge, these models could learn more robust and precise representations from the complex weather data. Besides, accelerating training and inference is important (Song et al., 2020) since diffusion models often incur high computational overheads.

5.4 Multi-Modal Learning

Weather data comes from heterogeneous sources, encompassing observational data (e.g., sensors, radar, satellite imagery), reanalysis data, and supplementary text descriptions (Li et al., 2024a).

Opportunities: These modalities can complement each other, offering a more comprehensive understanding of weather and climate phenomena. Therefore, a promising direction is to leverage such multi-modal data to learn joint representations of weather and climate events. However, a key challenge lies in effectively “aligning” these multi-modal data. Mapping numerical data to textual descriptions presents an additional layer of complexity. One possibility involves leveraging large language models (LLMs) to construct knowledge graphs that extract information about weather and climate events from corpora of environment-focused news articles. These extracted events can then be linked with meteorological raster data to

enrich the model’s understanding and predictive capabilities (Li et al., 2024a).

5.5 Data Processing and Management

Data Storage: The volume of weather and climate data is increasing daily - European Centre for Medium-Range Weather Forecasts (ECMWF) archives contain about 450 PB of data to which 300 TB are added daily (Mukkavilli et al., 2023).

Opportunities: Variational Autoencoder (VAE) approaches have emerged as powerful tools for data compression (Liu et al., 2024c; Han et al., 2024a), converting the high-dimensional data from the original space to a lower latent space. Liu et al. reduce the data size from 8.61 TB to a compact 204 GB and Han et al. compress the ERA5 dataset (226 TB) into a CRA5 dataset (0.7 TB). More importantly, they demonstrate that downstream experiments of global weather forecasting models trained on the compact CRA5 dataset achieve accuracy comparable to the models trained on the original dataset. This approach significantly reduces storage requirements for massive weather datasets.

Data Quality: Massive gridded reanalysis data are computed using mechanical or statistics models, which are still based on empirical assumptions. Thus, the quality of the reanalysis data is of concern.

Opportunities: Data assimilation (Manshausen et al., 2024) is a promising method to increase data quality by calibrating model outputs with observational data, which could be remote sensing imagery and ground station measurements. For example, SLAMS proposes a conditional diffusion model to assimilate *in situ* weather station data and *ex situ* satellite imagery to effectively calibrate the vertical temperature profiles (Qu et al., 2024), ADAF employs Swin Transformer to achieve effective data assimilation using real-world observations from different locations and multiple sources, including sparse surface weather observations and satellite imagery (Xiang et al., 2024). Furthermore, EarthNet is a multi-modal foundation model for global data assimilation of Earth observations utilizing masked autoencoders (Vandal et al., 2024). In summary, DL methods have become increasingly popular for integrating weather data from various sources to provide more precise representations.

6 Discussion

We have introduced three categories of models in Section 3. Each approach offers unique strengths

and trade-offs, making them suitable for different scenarios depending on the nature of the task, data availability, and computational resources. Below, we provide a detailed comparison and analysis of what works best in different scenarios, exploring why certain models excel in specific contexts.

Deterministic Predictive Models. These models have demonstrated exceptional performance for short-, medium- and long-range weather predictions. While Transformer-based models work well on temporal predictions, GNN-based models excel at modeling spatial relations, and hybrid models capture spatiotemporal dependencies with greater accuracies, but may require a longer time for training. WeatherBench 2 (Rasp et al., 2023) has benchmarked data-driven global medium-range (10 days) weather models and provides a detailed headline scorecard¹. In summary, NeuralGCM outperforms other state-of-the-art DL models, and it is comparable with the physics-based ECMWF’s IFS regarding geopotential, temperature, and wind variables. Models like GraphCast, Pangu, and FuXi have shown competitive or better performance compared with ECMWF’s High-RESolution forecast (HRES). However, three challenges remain. 1) Their output is usually blurry because they are typically trained to minimize a deterministic loss function that uses mean squared error (MSE). This becomes worse for extreme weather events. 2) They lack aleatoric and epistemic uncertainty quantification. Even though there have been attempts to use traditional initial condition perturbation methods to produce ensemble forecasts, modeling the uncertainty of weather evolution has not been addressed. 3) These models need architectural changes and re-training when applied to other specific tasks.

Probabilistic generative models. These models have shown great promise for accurate weather prediction. More importantly, probabilistic generative models such as GenCast, CoDiCast, and CasCast (see Figure 3) have brought unique strengths by modeling aleatoric and epistemic uncertainty due to the probabilistic noise sampling. These are particularly valuable for predicting extreme weather events, where probabilistic outputs can facilitate informed decision-making. GenCast has reported greater skill than IFS ENS on 97.4% of 1320 targets they evaluated. However, these models require more computational resources for training

and inference than deterministic predictive models, though they are faster than physics-based models.

Foundation models. Foundation models like Aurora, ClimaX and Prithvi WxC represent a significant leap in adaptability and transfer learning, offering robust performance across diverse tasks after fine-tuning. Furthermore, current foundation models are primarily based on deterministic predictive learning for pre-training, where latent embeddings are often obtained with predictive learning. We have not identified any that utilize probabilistic generative architectures. However, their large parameter size and pre-training requirements can create barriers for research groups with limited computational resources. Furthermore, fine-tuning techniques in weather forecasting are still in their early stages and could benefit from insights and advancements in the natural language processing domain (Zheng et al., 2023; Sun et al., 2022).

Table 2: Comparison of Predictive Learning, Generative Learning, and Pre-training & Fine-tuning Models for global medium-range (10 days) weather prediction.

	Predictive Learning	Generative Learning	Pre-training & Fine-tuning
Accuracy	NeuralGCM and FuXi are comparable with IFS ENS	GenCast: 97.4% targets better than IFS ENS	Aurora vastly better than IFS HERS
Efficiency	Fast training; Fast inference	Slow training; Slow inference	Slow training; Fast inference
Uncertainty	Need perturbation	Inherent	-
Adaptability	Need re-training	Need re-training	Fine-tuning

7 Conclusions

In this work, we present a comprehensive and up-to-date survey of data-driven deep learning models and foundation models for weather prediction. We introduce a novel categorization of these models based on their training paradigms and provide an in-depth review, analysis, and comparison of key methodologies within each category. Additionally, we summarize available datasets, open-source codebases, and diverse real-world applications in a GitHub repository. More importantly, we outline ten critical research directions across five primary avenues for advancing AI-driven weather prediction, offering a roadmap for future research.

Limitations. In this survey, we are particularly targeting the topic of weather prediction. Due to the limited space, other research topics in weather and climate domains are out of the scope of this survey, including climate downscaling (Ling et al., 2024a), climate emulation (Yu et al., 2024b), and climate trend prediction (Cael et al., 2023).

¹<https://sites.research.google/weatherbench/>

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Appendix

A Datasets

We summarize widely used benchmark datasets, where each data set is presented by domain, name, coverage, collection method, spatial and temporal resolution, time span, and the paper that introduces the dataset.

Table 3: Summary of Publicly Available Data Sets on Weather. CAM5: Community Atmospheric Model v5.

Domain	Dataset	Coverage	Collect	Spatial	Temporal	Time Span	Paper
General Weather	WeatherBench	Global	Reanalysis	1.40625°, 2.8125°, 5.625°	6 hours	1979-2018	(Rasp et al., 2020)
	WeatherBench 2	Global	Reanalysis	0.25°	6 hours	1979-2020	(Rasp et al., 2023)
	Weather2K	Region in China	Observation	-	1 hour	2017.01-2021.08	(Zhu et al., 2023)
	Weather5K	Global	Observation	-	1 hour	2014-2023	(Han et al., 2024b)
	HR-Extreme	Region in U.S.	Radar	3 km × 3 km	1 hour	2020-2020	(Ran et al., 2024)
Precipitation	SEVIR	Region in U.S.	Radar&Satellite	1 km × 1 km	5 mins	2017-2019	(Veillette et al., 2020)
	OPERA	Europe	Radar&Satellite	2 km	15 mins	2019-2021	(Herruzo et al., 2021)
	Metconet	France	Radar&Satellite	1 km	5-15 mins	2016-2018	(Larvor et al., 2020)
	IMERG	Global	Radar&Satellite	1 km	30 mins	2020-2023	(Huffman et al., 2020)
	HKO-7	Region in Hong Kong	Radar	1 km × 1 km	6 mins	2009-2015	(Shi et al., 2017)
	Shanghai	Shanghai	Radar	1 km	6 mins	2015-2018	(Chen et al., 2020)
	JMA	Japan	Radar	1 km	5 mins	2015-2017	(Inoue and Misumi, 2022)
	MRMS	CONUS and S. Canada	Radar	1 km × 1 km	2 mins	2017-2019	(Smith et al., 2016)
	RYDL	Germany	Radar	1 km	5 mins	2014-2015	(Ayzel et al., 2020a)
	RainBench	-	-	5.625°	1 hour	2016-2019	(de Witt et al., 2021)
	IowaRain	Iowa, U.S.	Radar	0.5 km × 0.5 km	5 mins	2016-2019	(Sit et al., 2021)
	PostRainBench	Region in China	Radar	1 km × 1 km	3 hours	2010-2021	(Tang et al., 2023)
Wind	GlobalWindTemp	Global	Observation	-	1 hour	2019-2020	(Wu et al., 2023)
	DigitalTyphoon	W.N. Pacific basin	Satellite	5 km	1 hour	1978-2022	(Kitamoto et al., 2023)
	TropicalCyclone	Global	CAM5 simulation	25 km	3 hours	1979-2005	(Racah et al., 2017)
	ClimateNet	Global	CAM5 simulation	25 km	3 hours	1996-2010	(Kashinath et al., 2021)
Air Quality	UrbanAir	Regional, China	Observation	-	1 hour	2014-2015	(Zheng et al., 2013)
	KnowAir	Regional, China	Observation	-	3 hours	2015-2018	(Wang et al., 2020)
	ItalianAir	Italy	Observation	-	1 hour	2004-2005	(Vito, 2016)
	BeijingAir1	Regional, China	Observation	-	1 hour	2010-2014	(Chen, 2017)
	BeijingAir2	Regional, China	Observation	-	1 hour	2013-2017	(Chen, 2019)
SST	OI SST v2	Pacific Ocean	Observation&Satellite	5°S-5°N, 170°W-120°W	Daily	1982-2017	(Huang et al., 2019)
	ZonalWinds	Pacific Ocean	Reanalysis	5°S-5°N, 120°E-160°E	Daily	1982-2017	(Huang et al., 2019)
	TropicalOcean	Pacific Ocean	Observation	5°S-5°N, 120°E-80°W	Monthly	1982-2017	(Huang et al., 2019)
	SODA SST	Global	Reanalysis	5° × 5°	Monthly	1871-1973	(Geng and Wang, 2021)
	GODAS	Global	Reanalysis	5° × 5°	Monthly	1994-2010	(Geng and Wang, 2021)
	CMIP5	Global	Simulation	5° × 5°	Monthly	1861-2004	(Geng and Wang, 2021)
	ERA-Interim	Global	Reanalysis	-	Monthly	1984-2017	(Ham et al., 2019)
	CFSv2	Global	Reanalysis	5° × 5°	6 hours	1981-2017	(He et al., 2019)
	NOAA ERSSTv5	Global	Observation	-	Monthly	1854-2020	(Cachay et al., 2020)
	CMIP6	Tropical Pacific	Simulation	2° × 0.5°	Monthly	1850-2014	(Zhou and Zhang, 2023)
	ORAS5	Tropical Pacific	Reanalysis	-	Monthly	1958-1979	(Zhou and Zhang, 2023)
	NOAA/CIRE	Global	Reanalysis	2° × 2°	6 hours	1850-2015	Mu et al.
	REMSS	Global	Satellite	0.25° × 0.25°	Daily	1997-2020	Mu et al.
	ENSO	Tropical Pacific	NOAA, NCEI, NCAR	-	Monthly	1950-2023	(Mir et al., 2024)
	GHRSSST	South China Sea	Observation	1.20° × 1.20°	Daily	2007-2014	(Meng et al., 2023)
	HYCOM	South China Sea	Simulation	1.12° × 1.12°	Daily	2007-2014	(Meng et al., 2023)
	Hadley-OI SST	Global	Observation&Satellite	1° × 1°	Monthly	1870-2020	(Liu et al., 2023b)
	COBE SST	Global	Observation	1° × 1°	Monthly	1891-2020	(Liu et al., 2023b)
	SILO SST	Australia	Observation	-	Monthly	1921-2020	(He et al., 2024b)
	OISST	Global	Observation&Reanalysis	0.25° × 0.25°	Daily	1982-2020	(He et al., 2024a)
	ERA5	Global	Observation&Reanalysis	0.25° × 0.25°	1 hour	1982-2020	(He et al., 2024a)
Flood	DEM	Carlisle, UK	Observation	5 m	1 hour	2005-2015	(Kabir et al., 2020)
	AustraliaFlood	Australia	Observation	-	Daily	1900-2018	(Adikari et al., 2021)
	SekongFlood	Vietnam, Laos, Cambodia	Observation	-	Daily	1981-2013	(Adikari et al., 2021)
	BangladeshFlood	Bangladesh (GBM river network)	Observation	-	Daily	1979-2014	(Ruma et al., 2023)
	GermanyFlood	Germany, Sachsen	Radar	1 km	1 hour	Different periods	(Li et al., 2022)
	ElbeRiverFlow	Germany, Elbe River in Sachsen	Observation	-	1 hour	Different periods	(Li et al., 2022)
	FijiFlood	Fiji Islands	Observation	-	Daily	1990-2019	(Moishin et al., 2021)
	FloridaFlood	USA, Coastal South Florida	Observation	-	1 hour	2010-2020	(Shi et al., 2024b)
	QijiangRiverBasin	China, Chongqing, Qijiang River	Observation	-	1 hour	1979-2020	(Shao et al., 2024)
	TunxiRiverBasin	China, Anhui, Tunxi River	Observation	-	1 hour	1981-2007	(Shao et al., 2024)
	MODIS	Regional, China	Satellite	500 m	Monthly	2000-2020	(Zhang et al., 2023a)
Drought	CHIRPS	Regional, China	Satellite	0.05°	Monthly	2000-2020	(Zhang et al., 2023a)
	ChinaDrought	China	-	-	Monthly	1980-2019	(Xu et al., 2022)
	IndianDrought	Peninsular, India	Satellite	0.25° × 0.25°	Daily	1981-2021	(Shukla and Pandya, 2023)
	AVHRR	Peninsular, India	Radiometer	1 km	Daily	1981-2022	(Shukla and Pandya, 2023)
	ERA5	East Asia	Reanalysis	0.25° × 0.25°	1 hour	1970-2020	(Zhang et al., 2024a)
	EastAsiaDrought1	East Asia	Satellite	0.25°	Daily	2003-2018	(Park et al., 2020)
	EastAsiaDrought2	East Asia	Satellite	0.05°	16 days	2003-2018	(Park et al., 2020)
	EastAsiaDrought3	East Asia	Satellite	0.05°	8 days	2003-2018	(Park et al., 2020)
	EastAsiaDrought4	East Asia	Simulation	0.5°	3 hours	2015-2018	(Park et al., 2020)
	EastAsiaDrought5	East Asia	Satellite	90 m	-	-	(Park et al., 2020)
	EastAsiaDrought6	East Asia	Satellite	0.5°	Yearly	-	(Park et al., 2020)
Wildfire	LANDFIRE PROGRAM	California	Satellite	128 × 128	15 mins	-	(Burge et al., 2023)
	FARSITE	Regional	Synthetic	30 m	15 mins	-	(Burge et al., 2023)
	NASA-MODIS Terra	California	Satellite	1 km	5 mins	2017-2018	(Chowdhury et al., 2021)
	MERRA-2	California	Reanalysis	0.5° × 0.625°	1 hour	2017-2018	(Chowdhury et al., 2021)
	USGS	Regional	Satellite	30 m	-	2017-2018	(Chowdhury et al., 2021)
	AICC	Regional, Alaska	Satellite	400 × 350	Daily	2002-2018	(Marjani et al., 2023)
	NRC	Regional, Canada	Satellite	30 m	Daily	2002-2018	(Marjani et al., 2023)
	VIIRS	South Africa	Satellite	375 m	1 hour	2012-2014	(Perumal and Van Zyl, 2020)
	VIIRS	California	Satellite	375 m	Daily	2012-2021	(Masrur et al., 2024)
	Percolation model	Regional	Synthetic	110 × 110	5 mins	-	(Masrur et al., 2024)

B Model Architectures

B.1 Convolutional Neural Networks

Convolutional Neural Networks (CNNs) (LeCun et al., 1995) are a specialized type of neural network designed for processing structured grid data, such as images. The convolutional layer usually utilizes

convolutional kernels to process the input data, performing convolution operations to extract features like edges, textures, and patterns (Li et al., 2021). This is often followed by a pooling layer to reduce the spatial dimensions of the feature maps, making the network computationally more efficient and focusing on the most important information.

They are widely used in tasks related to computer vision, such as image classification (He et al., 2016), object detection (Ren et al., 2016), and segmentation (He et al., 2017). Moreover, CNNs could be categorized into Conv1D, Conv2D, and Conv3D according to the sliding dimension of convolutional kernels (Kiranyaz et al., 2021).

B.2 Recurrent Neural Networks

Recurrent Neural Networks (RNNs) (Medsker and Jain, 2001) is a type of neural network particularly suited for tasks involving time-dependent or sequential data, such as time series forecasting (Sbrana et al., 2020), natural language processing (Mikolov et al., 2011; Zhang et al., 2017), and speech recognition (Yadav et al., 2022). The key idea behind this is to recurrently learn from a sequence of data with an internal (hidden) state, which includes as inputs the previous hidden states and current input. The learning or update rule is:

$$\begin{aligned} h_t &= \sigma(\mathbf{W}_x x_t + \mathbf{W}_h h_{t-1} + b_h), \\ y_t &= \sigma(\mathbf{W}_y h_t + b_y), \end{aligned} \quad (2)$$

where h_t is the hidden state at t -th time step, x_t is the input at t -th time step, y_t is the output at the same time step, $\mathbf{W}_x$, $\mathbf{W}_h$, and $\mathbf{W}_y$ are the weight matrices, b_h and b_y are the biases, and σ is the activation function (e.g., tanh or ReLU).

However, RNNs often suffer from gradient vanishing and gradient explosion while modeling long sequences. Long Short-Term Memory (Hochreiter and Schmidhuber, 1997) (LSTM) and Gated Recurrent Unit (Chung et al., 2014) (GRU) have been proposed to alleviate such a problem by well-designed gates to forget and filter information.

B.3 Graph Neural Networks

Graph Neural Networks (GNNs) (Scarselli et al., 2008) is designed to work on graph-structured data, $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, consisting of a set of nodes $\mathcal{V}$ and a set of edges $\mathcal{E}$. These nodes and edges represent the entities and the dependent relationships among these entities, respectively. Spatio-temporal Graph Neural Networks (ST-GNNs) (Yu et al., 2017) is an extension of GNNs designed to model both spatial and temporal dependencies in dynamic graph-structured data changing over time, $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}, t)$. Here, nodes $\mathcal{V}$ refer to spatial locations, and edges $\mathcal{E}$ refer to spatial relationships. Each node v_t^i represents the feature vector at the corresponding location i and time t . For each node, the message-passing technique (Gilmer et al., 2017) is often employed to capture the spatial dependencies on its neighbors. The temporal dependencies between graph snapshots can be modeled with the sequential models aforementioned. For the message passing, hidden states h_t^i at each node are updated based on messages (feature vectors) v_{t+1}^i according to:

$$\begin{aligned} v_{t+1}^i &= \sum_{j \in N(i)} M_t(h_t^i, h_t^j, e_{ij}), \\ h_{t+1}^i &= \sigma(h_t^i, v_{t+1}^i), \end{aligned} \quad (3)$$

where in the sum, $N(i)$ denotes the neighbors of i^{th} node in graph $\mathcal{G}$. After iterative updates k time steps, the final output of the whole graph at time $t + k$ can be computed with a readout function $\mathcal{O}$:

$$y_{t+k} = \mathcal{O}(\{h_{t+k}^i \mid i \in \mathcal{G}\}). \quad (4)$$

B.4 Transformer and Vision Transformer

To overcome the limitations of RNNs, which stem from their inherent sequential processing, the Transformer model (Vaswani, 2017) has emerged as a powerful alternative. Its core innovation lies in the use of parallel processing through the *attention* mechanism, enabling it to capture dependencies between any

parts of a sequence without the need for sequential steps (Wen et al., 2022). The *attention* mechanism is described as follows:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right) \mathbf{V}, \quad (5)$$

where the d_k denotes the dimension of the key, $\mathbf{Q} \in \mathbb{R}^{n \times d_k}$, $\mathbf{K} \in \mathbb{R}^{m \times d_k}$, and $\mathbf{V} \in \mathbb{R}^{m \times d_v}$ are the query matrix, key matrix, and value matrix, respectively. These three matrices are computed by linear transformations from the original input sequence $\mathbf{X} \in \mathbb{R}^{n \times d}$ with learnable weight matrices $\mathbf{W}_q \in \mathbb{R}^{d \times d_k}$, $\mathbf{W}_k \in \mathbb{R}^{d \times d_k}$, $\mathbf{W}_v \in \mathbb{R}^{d \times d_v}$, as

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_q, \mathbf{K} = \mathbf{X}\mathbf{W}_k, \mathbf{V} = \mathbf{X}\mathbf{W}_v. \quad (6)$$

Vision Transformer. The Vanilla Transformer was originally proposed for dealing with sequences. Vision Transformer (ViT) (Dosovitskiy et al., 2020) is a variant tailed to process images and has shown powerful performance compared to convolutional neural networks (CNNs). ViT models divide the input image into a grid of smaller, non-overlapping patches. Each patch is treated similarly to a “word” in natural language processing, and the patches are then flattened into vectors. Positional embeddings are added to these patch embeddings to mark the relative positions of patches in the image, helping models understand the image’s spatial layout. Subsequently, the additive embeddings are fed into the Vanilla Transformer layer to leverage the *attention* mechanism. We refer readers to look into Figure 1 in (Dosovitskiy et al., 2020).

B.5 Mamba and Vision Mamba

We start by introducing the State Space Models (SSMs). SSMs represent the evolution of the system’s internal states and make predictions of what their next state could be. For sequence modeling, SSMs map a sequence $x(t) \in \mathbb{R}^L \mapsto y(t) \in \mathbb{R}^L$ through an implicit latent state $h(t) \in \mathbb{R}^{L \times N}$:

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t), \\ y(t) &= \mathbf{C}h(t), \end{aligned} \quad (7)$$

where $\mathbf{A} \in \mathbb{R}^{N \times N}$ and $\mathbf{B}, \mathbf{C} \in \mathbb{R}^{N \times 1}$ are learnable matrices. The continuous sequence is discretized by a step size Δ , and the discretized SSM model is represented as:

$$\begin{aligned} h_t &= \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t, \\ y_t &= \mathbf{C}h_t, \end{aligned} \quad (8)$$

where discretization rule can be achieved by zero-order hold (Zhang and Chong, 2007) $\bar{\mathbf{A}} = \exp(\Delta\mathbf{A})$ and $\bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B}$. The structured state-space model (S4), a variant of the vanilla SSM, improves long-range dependency modeling by utilizing the High-order Polynomial Projection Operators (HiPPO) (Gu et al., 2020).

Mamba. S4 applies the same parameters $\mathbf{A}$ and $\mathbf{B}$ to each “token” of input, which is challenging to identify the importance of each input. Selective State Space Model (Mamba) (Gu and Dao, 2023) incorporates a selection mechanism such that parameters that affect interactions along the sequence are input-dependent (parameters Δ , $\mathbf{A}$, $\mathbf{B}$ are functions of the input), enabling capturing contextual information in long sequences. Besides, Mamba possesses efficient hardware-aware designs. It utilizes three computing acceleration techniques (kernel fusion, parallel scan, and recomputation) to materialize the hidden state h only in more efficient levels of the GPU memory hierarchy.

Vision Mamba. Vision Mamba (Zhu et al., 2024a) is a variant of Mamba used for image modeling. Similar to Vision Transformer, Vision Mamba first splits the input image into patches and then projects them into patch tokens, but leverages bidirectional SSMs (Mamba blocks) to replace attention mechanisms as the image encoder to model the sequence of tokens. Therefore, Vision Mamba can be well-tailed for 2-D grid weather data, e.g., MetMamba (Qin et al., 2024).

B.6 Generative Adversarial Networks

Generative Adversarial Networks (GANs) (Goodfellow et al., 2014; Mirza, 2014) were originally proposed to learn a generative model to generate realistic images via adversarial training. Specifically, GANs simultaneously train two neural networks adversarially: a Generator G and a Discriminator D . The Generator learns the underlying data distribution and generates produce samples that can effectively fool the discriminator, while the discriminator differentiates between the samples generated by the generator and the real samples by outputting the corresponding probabilities. This training process can be regarded as a two-player zero-sum game (Washburn and Wood, 1995), ultimately ending when the discriminator is unable to distinguish between the generator-generated samples and the real samples, i.e., $D(x) = \frac{1}{2}$.

GANs have widely used for image generation (Xu et al., 2018), super-resolution (Harder et al., 2022), style transferring (Zheng et al., 2022), and image-based weather forecasting (Chen et al., 2022; Choi et al., 2023; Cheng et al., 2023).

B.7 Diffusion Models

Diffusion Models (DMs) (Ho et al., 2020; Song et al., 2020) are the other type of generative models that have gained significant popularity in computer vision (Saharia et al., 2022; Croitoru et al., 2023), natural language processing (Hertz et al., 2022; Li et al., 2023b), due to their ability to produce high-quality, realistic samples. Diffusion models work in two processes: *forward diffusion process* and *reverse denoising process*. In the forward process, data (e.g., an image) is gradually “noised” by adding small amounts of Gaussian noise over multiple steps until it becomes nearly pure noise. This process is usually fixed and non-learnable, where each step incrementally increases the noise. The reverse process is learnable, where the model learns how to gradually remove noise, step-by-step, to recover a realistic sample from a noisy starting point. This iterative denoising process helps to learn the intricate, high-dimensional data distribution.

Mathematically, the *forward process* transforms an input $\mathbf{x}_0$ with a data distribution of $q(\mathbf{x}_0)$ to a white Gaussian noise vector $\mathbf{x}_N$ in N diffusion steps. It can be described as a Markov chain that gradually adds Gaussian noise to the input according to a variance schedule $\{\beta_1, \dots, \beta_N\} \in (0, 1)$:

$$q(\mathbf{x}_{1:N} | \mathbf{x}_0) = \prod_{n=1}^N q(\mathbf{x}_n | \mathbf{x}_{n-1}), \quad (9)$$

where at each step $n \in [1, N]$, the diffused sample $\mathbf{x}_n$ is obtained with $q(\mathbf{x}_n | \mathbf{x}_{n-1}) = \mathcal{N}(\mathbf{x}_n; \sqrt{1 - \beta_n} \mathbf{x}_{n-1}, \beta_n \mathbf{I})$.

In the *reverse process*, the *denoiser network*, $p_\theta(\cdot)$, is used to recover $\mathbf{x}_0$ by gradually denoising $\mathbf{x}_n$ starting from a Gaussian noise $\mathbf{x}_N$ sampled from $\mathcal{N}(0, \mathbf{I})$. This process is presented as:

$$p_\theta(\mathbf{x}_{0:N}) = p(\mathbf{x}_N) \prod_{n=1}^N p_\theta(\mathbf{x}_{n-1} | \mathbf{x}_n). \quad (10)$$

In weather and climate domains, diffusion models have been applied to precipitation nowcasting (Asperti et al., 2023a; Gao et al., 2024), atmospheric downscaling (Ling et al., 2024a; Mardani et al., 2023), weather forecasting (Shi et al., 2024a; Andrae et al., 2024).